Reinforcement Learning for Optimal Setpoint Determination in Energy Utility Systems

Master Thesis

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Task description

Energy utility systems are the backbone of modern energy infrastructure. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimization. These approaches rely on accurate system models and are typically formulated as constrained optimization problems. While effective for well-defined and moderately sized systems, classical optimization becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from the integration of renewable sources, and model-plant mismatches makes it challenging to use purely model-based approaches.

This thesis investigates whether reinforcement learning can find optimal operating setpoints for two case studies of different complexity: a battery energy storage system and a non-linear cookie production factory. For each system, the control task is formulated as a Markov decision process (MDP) with explicit state, action, and reward definitions aligned with operational objectives. The RL agent is trained to produce adaptive setpoint policies that minimize operating cost while satisfying process constraints and preserving stable operation. The learned policies are then benchmarked against established mathematical optimization methods to determine where RL provides practical benefits and where its limitations remain.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including total operational cost, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimization and will determine under which conditions RL offers practical advantages for real-time setpoint optimization in complex energy utility systems.

Abstract

Example summary in English.

Zusammenfassung

Beispielzusammenfassung in Deutsch.

Contents

1 Introduction

Introduces the research problem, motivation, and objectives of the thesis.

2 Theory and Background

Provides the fundamental concepts and background required to understand the systems, optimization methods, and reinforcement learning used in this work.

This section provides the theoretical basis for the control problem studied in this thesis. The central question is how operating setpoints in energy utility systems can be selected over time so that costs are reduced while physical and operational limits are respected. The same problem can be viewed from two perspectives: as a model-based constrained optimization problem, and as a sequential decision-making problem that can be addressed with reinforcement learning. The following sections introduce both perspectives and then position Soft Actor-Critic (SAC), the reinforcement-learning algorithm used in this work.

2.1 Setpoint Optimization in Energy Utility Systems

Defines setpoints and explains why their optimal selection is relevant for energy utility systems.

Energy utility systems are operated by choosing setpoints for controllable components such as storage units, heat pumps, pumps, valves, and bypasses. A setpoint describes the desired value of a control variable. In the systems considered here, this may be the charging or discharging power of a battery, the compressor setting of a heat pump, a mass flow rate, or a temperature level in a thermal process. These choices are not merely technical details; they directly affect operating cost, energy efficiency, constraint satisfaction, and system stability.

Setpoint determination is also inherently sequential. The best operating point at one time step cannot be chosen independently of what happened before or what may be needed later. Storage systems make this especially clear. Charging a battery or a thermal energy storage can increase future flexibility, but it may also increase the current electricity cost. Discharging can reduce current grid consumption, but it leaves less stored energy for later. Good setpoint decisions therefore have to account for inter-temporal dependencies, changing boundary conditions, and physical operating limits at the same time.

In practical operation, the controller also has to react to external signals such as electricity prices, renewable generation, production demand, and ambient conditions. Since these signals change over time, fixed rule-based strategies can become inefficient. The control task in this thesis is therefore to determine feasible setpoints over a finite time horizon while minimizing operating cost and respecting constraints such as power limits, temperature bounds, storage capacities, and process requirements. This problem is studied for two systems of different complexity: a battery energy storage system and an electrified steam-generation system for a cookie production factory.

2.2 Mathematical Optimization

Presents traditional optimization methods used for determining optimal system setpoints.

Mathematical optimization provides a natural way to describe setpoint selection. In general, optimization means selecting the best decision from a set of feasible alternatives according to a defined objective. For energy systems, the objective is often to minimize operating cost or energy consumption, while the feasible set is determined by physical equations and operational limits. A constrained optimization problem can be written as [1]:

$$\min_{x\in\Omega} f(x) \quad \text{s.t.} \quad h(x) = 0,\ g(x) \leq 0, \tag{1}$$

Here, $x \in \Omega \subseteq \mathbb{R}^n$ denotes the decision variables, $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, $h : \mathbb{R}^n \to \mathbb{R}^p$ represents equality constraints, and $g : \mathbb{R}^n \to \mathbb{R}^q$ represents inequality constraints. The solution $x^\star$ is optimal if it minimizes the objective while satisfying all constraints.

For finite-horizon setpoint optimization, the decision vector contains the control actions for several time steps. The objective is usually the accumulated operating cost over the horizon. The constraints describe system dynamics, energy balances, storage bounds, and terminal requirements. This time-coupled structure is important because a state such as battery state of charge or thermal storage temperature depends on previous decisions.

The difficulty of an optimization problem depends strongly on its mathematical structure. Linear and mixed-integer linear formulations are widely used in energy-system applications because simplified models can often be solved reliably. However, realistic energy conversion systems frequently include nonlinear component behavior, continuous control variables, and coupled dynamic constraints. Such systems lead to nonlinear programming (NLP) formulations, which are generally harder to solve and may contain local optima [1].

Another important distinction is whether the problem is deterministic or stochastic. In deterministic optimization, all relevant parameters, such as electricity prices, wind availability, and demand, are assumed to be known over the optimization horizon. Stochastic optimization represents uncertain quantities through random variables or scenarios. This is closer to real operation, but it also increases model size and computational complexity.

Nonlinear constrained problems are usually solved with dedicated numerical solvers. Interior-point methods are common in this setting. IPOPT, for example, is an open-source solver for large-scale nonlinear programming and uses a primal-dual interior-point method with a filter line-search strategy [2]. These solvers can be very effective when the model is well defined. At the same time, their performance depends on model accuracy, constraint structure, scaling, and initialization. For nonlinear and nonconvex energy systems, solution times can increase, and the result may depend on the starting point. This can make direct real-time use difficult [3].

In this thesis, mathematical optimization serves as the model-based reference approach. Reinforcement learning is then investigated as an alternative that learns a policy for mapping observed system states directly to control actions after training.

2.3 Reinforcement Learning

Introduces reinforcement learning as a data-driven approach for learning optimal control policies through interaction with the environment.

Reinforcement learning (RL) offers a different view of the same setpoint problem. Instead of solving an optimization problem from scratch at every decision time, an agent learns from repeated interaction with an environment. At each time step, the agent observes the current state, selects an action, receives a reward, and observes the next state. Over time, it adapts its policy so that actions leading to high long-term reward become more likely [4].

Formally, RL problems are commonly modeled as a Markov Decision Process (MDP), defined by the tuple

$$(\mathcal{S}, \mathcal{A}, p, r, \gamma_{\mathrm{RL}}),$$

where

- $\mathcal{S}$ is the set of states representing the system,
- $\mathcal{A}$ is the set of actions,
- $p(s' \mid s, a)$ defines the transition dynamics,
- $r(s, a)$ is the reward function,
- $\gamma_{\mathrm{RL}} \in [0, 1]$ is the discount factor.

The Markov property means that the current state contains the information needed to describe the probability distribution of the next state. A policy $\pi(a \mid s)$ describes how the agent selects actions. The return from time step k is the discounted cumulative reward

$$G_k = \sum_{i=0}^{N-k-1} \gamma_{\mathrm{RL}}^{i} r(s_{k+i}, a_{k+i}),$$

and the learning objective is to find a policy that maximizes the expected return. Value functions estimate the expected return from a state, while action-value functions estimate the expected return from a state-action pair. These estimates are important because they allow the agent to consider the future consequences of a decision, not only its immediate reward [4].

In energy-system applications, the decision problem can often be interpreted as stochastic optimal control. Uncertainty enters through external variables such as renewable generation, electricity prices, and demand. These variables can be modeled as stochastic processes, which leads to a finite-horizon MDP formulation [5]. Since the engineering objective is usually cost minimization, the reward is often defined as the negative operating cost. The resulting goal is to minimize expected operating cost over time while respecting system dynamics and constraints.

Classical dynamic programming can solve some MDPs through backward recursion. However, this approach becomes impractical when the state or action space grows, a difficulty known as the curse of dimensionality. Realistic energy systems often have continuous states, continuous actions, and long horizons, so exact dynamic-programming solutions are usually not tractable.

RL addresses this limitation by learning approximate policies from interaction data. A trained policy can then compute an action directly from the current observation. In model-free RL, the learning algorithm does not require an explicit analytical transition model for planning, although training still requires interaction data. In engineering applications, this data is usually generated with a simulator or digital model, so the quality of the learned policy depends strongly on the quality of the training environment.

To locate the algorithm used in this thesis, it is helpful to briefly classify the main families of RL algorithms. RL methods are not separated by one single list of types; instead, they are usually described through several overlapping distinctions [4]:

- Model-based and model-free methods: model-based methods learn or use a model of the environment dynamics for planning. Model-free methods learn directly from sampled transitions.

- Value-based and policy-based methods: value-based methods learn value or action-value functions and choose actions from them. Policy-based methods directly optimize a parameterized policy.

- Actor-critic methods: actor-critic methods combine both ideas. The actor represents the policy, and the critic estimates values or action-values used to improve that policy.

- On-policy and off-policy methods: on-policy methods learn from data generated by the current policy. Off-policy methods can also reuse data generated by previous policies or exploratory behavior.

- Tabular and deep RL methods: tabular methods store values explicitly for small state and action spaces. Deep RL uses neural networks as function approximators and is therefore more suitable for large or continuous spaces.

The choice of algorithm follows from the structure of the problem. In this thesis, the relevant setpoints are continuous quantities, such as battery power, temperature levels, mass flow rates, and bypass positions. Discretizing these variables would either remove useful operating points or create a very large action space. A suitable algorithm should therefore learn a parameterized policy and output continuous actions directly.

Policy-gradient and actor-critic methods are commonly used for such problems because they optimize a parameterized policy instead of enumerating all possible actions [5]. In an actor-critic setting, the actor can propose continuous setpoints, while the critic evaluates their expected long-term effect on cost and future system states. Within the classification above, Soft Actor-Critic (SAC) is a model-free, off-policy, actor-critic, deep reinforcement-learning algorithm.

SAC is used in this thesis because it extends the model-free actor-critic framework with a maximum-entropy objective for continuous-control problems [6]. This is useful for the present application because the agent must search through continuous operating setpoints during training, while still learning a policy that can be evaluated quickly after training.

In standard RL, the policy is usually trained to maximize expected return. SAC adds an entropy term to this objective. Intuitively, this means that the policy is encouraged to keep exploring when several actions appear useful, instead of becoming too deterministic too early. The objective can be interpreted as maximizing both reward and stochasticity:

$$\mathbb{E}_\pi \left[\sum_{k=0}^{N-1} \gamma_{\mathrm{RL}}^k \left(r(s_k, a_k) + \alpha \mathcal{H}(\pi(\cdot \mid s_k)) \right) \right],$$

where α controls the weight of the entropy term and $\mathcal{H}$ denotes policy entropy [6]. This entropy term can improve exploration and reduce the risk that the agent converges too early to a narrow operating strategy.

SAC is therefore a suitable choice for the systems considered in this thesis. It can handle continuous action spaces, reuse past experience through off-policy learning, and support exploration in environments with time-varying prices, renewable generation, storage dynamics, and operational constraints.

3 Use Cases

Describes the two systems considered in this thesis and their operational challenges.

3.1 Battery Energy Storage System

Describes the battery system used as the first case study.

Battery energy storage systems (BESS) are electrochemical devices that store electrical energy for later use and provide grid services such as energy arbitrage, peak shaving, and short-term balancing. A typical BESS is characterized by its usable energy capacity, maximum charge/discharge power, round-trip efficiency, and state-of-charge (SOC) operating limits. Real-world operation must balance economic objectives (e.g., buying low / using stored energy when prices are high) with engineering constraints (SOC bounds, power limits, efficiency losses).

BESS presents several challenges. Decisions are inherently time-coupled because current charging/discharging changes future availability. In addition, uncertain exogenous signals (market prices) require policies that are robust to forecast errors, and hard feasibility constraints must be enforced at every time step. These properties make BESS an illustrative case study for comparing optimization-based and reinforcement-learning approaches to real-time setpoint determination in energy systems.

A battery energy storage system provides flexible, time-shiftable power by charging when electricity prices are low and discharging when prices are high to reduce grid procurement cost. In this thesis, the battery is modeled using the `BatteryEnv` environment: the agent's observation includes the state-of-charge (SOC), a normalized current price, a normalized time index, and a forecast window of upcoming prices; the action is a continuous charge/discharge command $a \in [-1, 1]$ scaled to real power $P \in [-P_{\max}, P_{\max}]$.

The SOC dynamics are modeled as

$$\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \frac{P_{\mathrm{actual},t}\,\Delta t}{E_{\max}},$$

where $P_{\mathrm{actual},t}$ is the effective storage-side power after efficiency losses, $E_{\max}$ the usable energy capacity, and Δt the time step. The implemented environment uses the sign convention $P_t > 0$ for charging and $P_t < 0$ for discharging, with a simplified symmetric efficiency coefficient η. The system is subject to the following operational constraints:

$$\begin{aligned} 0 \le \mathrm{SOC}_t \le 1, &\quad \forall t, && (2)\\ -P_{\max} \le P_t \le P_{\max}, &\quad \forall t, && (3)\\ P_{\mathrm{actual},t} = \eta P_t, &\quad \forall t. && (4) \end{aligned}$$

where $P_{\max}$ is the maximum charge/discharge power. Constraint (2) ensures the battery operates within safe operating levels, Constraint (3) enforces physical power limitations, and Constraint (4) models the efficiency losses used in the RL environment. The environment enforces feasibility by dynamically limiting the commanded power at each step so that the SOC remains within bounds.

The goal is to reduce the cost of electricity from the grid, which is calculated as the price multiplied by the power used. The environment gives the negative of this cost as the reward. In this work, there is no on-site generation and no selling back to the grid. The battery charges from the grid when prices are low and discharges to serve local demand when prices are high. Main challenges include respecting SOC and power limits, accounting for efficiency losses, handling uncertain future prices, and managing the fact that actions now affect future energy availability.

These features make the BESS a suitable test case for reinforcement learning, where a policy must balance using energy now against saving it for likely higher-price periods while always obeying feasibility limits.

3.2 Cookie Production Factory

Introduces the industrial production system considered as the second case study.

The second use case represents a non-linear cookie production factory with an electrified steam-generation utility system. The production process requires a constant supply of process heat in the form of superheated steam, while the utility system must decide how to operate the energy conversion components under fluctuating wind generation and electricity prices. The operational objective is to satisfy the steam demand while minimizing grid electricity cost over a finite scheduling horizon. The system model is based on the industrial power-to-heat system introduced by Walden et al. [7] and later used as a renewable steam-generation benchmark by Kyriakidis et al. [3].

The system consists of four main units: a wind turbine (WT), a closed reverse Brayton-cycle high-temperature heat pump (HTHP), a sensible thermal energy storage (TES), and a steam generator (SG). The WT supplies renewable electricity, while additional electricity can be purchased from the grid. The HTHP uses a waste-heat air stream as heat source and delivers high-temperature process heat to an intermediate thermal-oil loop. The thermal oil acts as heat transfer fluid and can be routed through the TES or directly to the SG by controllable bypasses. The SG provides the required steam conditions for the consumer factory. The HTHP and SG are represented by polynomial surrogate models, while the TES is modeled as a simplified single energy reservoir, and the charging/discharging behavior is represented using efficiency equations.

Placeholder for schematic of the electrified steam-generation system

Figure 1: Schematic representation of the electrified steam-generation system serving the cookie production factory, adapted from the industrial power-to-heat system introduced by Walden et al. [7] and the renewable steam-generation benchmark described by Kyriakidis et al. [3].

The corresponding finite-horizon optimization problem is formulated over a discrete time horizon with time index $k \in \{1, \ldots, n\}$:

$$
\begin{aligned}
\min_{\mathbf{x}} \quad & J = \sum_{k=1}^{n} P_{\mathrm{grid}}^{k} g_{\mathrm{grid}}^{k} \Delta t, && (5)\\
\text{s.t.} \quad & P_{\mathrm{grid}}^{k} + P_{\mathrm{WT}}^{k} = 3F_{\mathrm{HTHP}}\left(T_{\mathrm{I}}^{k}, \dot{m}_{\mathrm{I}}^{k}, T_{\mathrm{III}}^{k}, R^{k}\right), && (6)\\
& T_{\mathrm{II}}^{k} = F_{\mathrm{HTHX}}\left(T_{\mathrm{I}}^{k}, \dot{m}_{\mathrm{I}}^{k}, T_{\mathrm{III}}^{k}, R^{k}\right), && (7)\\
& T_{\mathrm{IV}}^{k} = F_{\mathrm{LTHX}}\left(T_{\mathrm{I}}^{k}, \dot{m}_{\mathrm{I}}^{k}, T_{\mathrm{III}}^{k}, R^{k}\right), && (8)\\
& T_{2}^{k} = \beta_{1}^{k} T_{\mathrm{II}}^{k} + \left(1 - \beta_{1}^{k}\right) T_{1}^{k}, && (9)\\
& T_{\mathrm{I}}^{k} = \beta_{2}^{k} T_{3}^{k} + \left(1 - \beta_{2}^{k}\right) T_{4}^{k}, && (10)\\
& T_{2}^{k} = 201.92 + \frac{1819.32}{3\dot{m}_{\mathrm{II}}^{k}}, && (11)\\
& T_{3}^{k} = 196.3 - \frac{188.4}{3\dot{m}_{\mathrm{I}}^{k}}, && (12)\\
& \dot{Q}_{s,\mathrm{ch}}^{k} = 3\dot{m}_{\mathrm{II}}^{k} c_{p,\mathrm{f}} \left(T_{\mathrm{II}}^{k} - T_{1}^{k}\right)\left(1 - \beta_{1}^{k}\right), && (13)\\
& \dot{Q}_{s,\mathrm{dch}}^{k} = 3\dot{m}_{\mathrm{I}}^{k} c_{p,\mathrm{f}} \left(T_{4}^{k} - T_{3}^{k}\right)\left(1 - \beta_{2}^{k}\right), && (14)\\
& \dot{Q}_{s,\mathrm{ch}}^{k} \dot{Q}_{s,\mathrm{dch}}^{k} \le \gamma, && (15)\\
& T_{1}^{k} = T_{\mathrm{II}}^{k} - \epsilon_{\mathrm{ch}}\left(T_{\mathrm{II}}^{k} - T_{s}^{k-1}\right), && (16)\\
& T_{4}^{k} = T_{3}^{k} - \epsilon_{\mathrm{dch}}\left(T_{3}^{k} - T_{s}^{k-1}\right), && (17)\\
& T_{s}^{k} = T_{s}^{k-1} + \frac{\dot{Q}_{s,\mathrm{ch}}^{k} - \dot{Q}_{s,\mathrm{dch}}^{k}}{m_{s} c_{p,\mathrm{s}}} \Delta t, && (18)\\
& T_{s}^{0} = \tilde{T}_{0}, \qquad T_{s}^{n} = T_{s}^{0}, && (19)\\
& T_{\mathrm{I}}^{k} \in [177, 250], \qquad \dot{m}_{\mathrm{I}}^{k} \in [5, 16], && (20)\\
& T_{\mathrm{III}}^{k} \in [60, 100], \qquad R^{k} \in [0.8, 1.53], \qquad \beta_{1}^{k}, \beta_{2}^{k} \in [0, 1]. && (21)
\end{aligned}
$$

Here, the objective in Eq. (5) minimizes the electricity cost of grid power consumption. The power balance in Eq. (6) couples grid electricity, wind power, and HTHP demand. The remaining constraints describe the HTHP and SG surrogate models, bypass mixing, TES charging and discharging, the prohibition of simultaneous charge and discharge, cyclic storage operation, and the admissible operating ranges of the main decision variables.

The main control decisions are the bypass variables β_1 and β_2, the compressor-speed-related variable R, the mass flow rate $\dot{m}_{\mathrm{I}}$, and the HTHP inlet temperature T_{I}. The bypass variables determine the operating mode of the system. In charging mode, heat from the HTHP is routed into the TES; in discharging mode, stored heat is released to support the steam-generation process; and in idle or bypass operation, the storage is not actively charged or discharged. Simultaneous charging and discharging is prohibited through a complementarity constraint.

In the reinforcement-learning environment, the agent observes the normalized TES temperature, current electricity price, available wind power, the previous operating mode, and forecast information for price and wind availability. The action is a continuous vector containing β_1, β_2, R, $\dot{m}_{\mathrm{I}}$, and T_{I}. The reward is based on the negative grid-electricity cost, with additional penalties for violating operational requirements such as TES temperature bounds, invalid mode behavior, coupling residuals, and deviation from the cyclic terminal storage condition.

This use case is more complex than the battery system because it combines nonlinear surrogate equations, thermal storage dynamics, multiple coupled continuous control variables, and hard process constraints. It therefore provides a suitable second benchmark for evaluating whether reinforcement learning can learn feasible and cost-effective setpoint policies for industrial energy utility systems, and for comparing the learned policy against the existing mathematical optimization formulation.

4 Methodology

Describes how the optimization problem is formulated as a reinforcement learning problem.

The previous section introduced the two energy utility systems considered in this thesis and described their operational objectives and constraints. This section explains how these system-specific optimization problems are reformulated as reinforcement-learning problems. In particular, the setpoint optimization task is expressed as a finite-horizon sequential decision problem, where an agent observes the current system state, selects continuous control actions, and receives a reward based on operating cost and constraint satisfaction.

4.1 Problem Formulation

Formulates the system control problem as a reinforcement learning problem.

For both use cases, the objective is to determine operating setpoints over a discrete time horizon such that the total operating cost is minimized while all relevant constraints are respected. Let $k \in \{0, \ldots, N-1\}$ denote the discrete time index. At each time step, the system is described by a state vector s_k, the controller selects an action vector a_k, and the system evolves to the next state s_{k+1} according to the underlying system dynamics and exogenous inputs such as electricity prices and renewable generation.

The resulting decision problem is formulated as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma_{\mathrm{RL}}),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $p(s_{k+1} \mid s_k, a_k)$ denotes the transition dynamics, $r(s_k, a_k)$ is the reward function, and γ_{RL} is the discount factor. The agent aims to learn a policy $\pi(a_k \mid s_k)$ that maximizes the expected cumulative reward [4]:

$$\max_{\pi} \; \mathbb{E}_{\pi}\left[\sum_{k=0}^{N-1} \gamma_{\mathrm{RL}}^{k} r(s_k, a_k)\right].$$

Since the original engineering objective is cost minimization, the reward is defined as the negative operating cost, augmented by penalty terms for constraint violations where required. In this way, feasible and economically favorable operation corresponds to a high cumulative reward. The following subsections describe how this general formulation is implemented in the reinforcement-learning framework, training setup, and evaluation procedure.

4.2 Reinforcement Learning Framework

Explains the structure of the RL agent and learning algorithm.

4.3 Training Setup

Describes the training environment, reward function, and hyperparameter configuration.

4.4 Evaluation Metrics

Defines the metrics used to evaluate the performance of the RL agent and optimization baseline.

5 Results and Discussion

Presents and analyzes the experimental results.

5.1 Battery Energy Storage System

5.2 Cookie Production Factory

5.3 Key Findings

6 Conclusion and Future Work

Summarizes the main findings and outlines possible future work.

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